

Bangla CQ Automated Evaluation Framework

A Human-Centered AI Rubric & Teacher Mental Model Architecture

Domain: Automated Essay & CQ Scoring Target Language: Bangla (সৃজনশীল প্রশ্ন) Methodology: Qualitative Grounded HCI

1. Introduction & Theoretical Foundation

Evaluating subjective responses in non-STEM disciplines like Bangla Creative Questions (সৃজনশীল প্রশ্ন - CQ) requires moving beyond rigid keyword-matching algorithms or binary decision trees. Human examiners rely on a dynamic **cognitive mental model** that balances explicit structural guidelines provided by secondary education boards with implicit qualitative judgments regarding handwriting, coherence, and language proficiency.

Core HCI Objective

The objective of this framework is not to construct conditional heuristics, but to encode the multi-tiered mental representation of an experienced evaluator into an AI scoring model that replicates human fairness, contextual flexibility, and domain-specific rigor.

2. Teacher Cognitive Workflow (The 3-Layer Evaluation Paradigm)

Qualitative analysis of teacher interview data reveals that an examiner processes a student script through three distinct sequential layers:

LAYER 1

Visual & Presentation Scan (The 5-Second Anchor)

Before reading the semantic content, the teacher forms an initial perceptual anchor based on script "outlooking." Key attributes evaluated include:

- **Visual Cleanliness (পরিচ্ছন্নতা):** Absence of excessive overwriting, scratch-outs (কাটাকাটি), and messy layout.
- **Structural Chunking:** Distinct visual separation of sub-questions ('ক', 'খ', 'গ', 'ঘ') using structural line-breaks or paragraphs.
- **Legibility vs. Calligraphy:** Handwriting style is not penalized if legible; however, visual clarity directly impacts reader cognitive load and initial impression.

LAYER 2

Structural Parsing & Cognitive Layer Matching

Teachers verify whether the response satisfies the mandatory cognitive layers established by the National Curriculum and Textbook Board (NCTB):

- **Part 'ক' (Knowledge / জ্ঞানমূলক - 1 Mark):** Binary verification of factual recall.
- **Part 'খ' (Comprehension / অনুধাবনমূলক - 2 Marks):** 2-part structure (জ্ঞান অংশ headline + অনুধাবন অংশ concise explanation).
- **Part 'গ' (Application / প্রয়োগমূলক - 3 Marks):** 3-part structure (Knowledge identification + Textual context + Stem-Text synthesis).
- **Part 'ঘ' (Higher Ability / উচ্চতর দক্ষতা - 4 Marks):** 4-part structure (Stance + Text analysis + Comparative alignment + Critical synthesis).

LAYER 3

Qualitative & Synthesis Calibration

The final score is fine-tuned by assessing language depth (ভাষাজ্ঞান), depth of explanation, avoidance of redundant stem copying, and analytical maturity in the student's own words.

3. Comprehensive Evaluation Matrix & AI Prompt Instructions

Part 'ক' — Knowledge (জ্ঞানমূলক - 1 Mark)

Cognitive Goal: Direct, accurate recall of literary facts, character names, or historical contexts.

Score	Evaluation Criteria	Teacher Mental Model & AI Instruction
1.0	Direct hit. Accurate fact/event provided.	Exact Recall: Student demonstrates direct knowledge of the text. Award full mark immediately regardless of length.
0.0	Incorrect, missing, or irrelevant fact.	Binary Gate: Incorrect answer flags superficial reading/preparation. Award 0.

Part 'খ' — Comprehension (অনুধাবনমূলক - 2 Marks)

Cognitive Goal: Stating the core reason upfront, followed by a concise contextual explanation (Target: 2 distinct paragraphs/sections).

Score	Evaluation Criteria	Teacher Mental Model & AI Instruction
2.0	Core headline sentence (জ্ঞান অংশ) + precise contextual explanation (অনুধাবন অংশ).	Optimal Response: States the main point clearly like a news headline, then provides a focused explanation in good literary tone.
1.0	Valid explanation present, but missing the explicit headline answer OR simply rephrases the prompt line.	Partial / Board Leniency: Student understands context but fails to formulate the precise headline reason. In school settings this is docked; under board settings, 1 mark is granted.
0.0	Off-topic text, missing response, or complete misinterpretation.	No Comprehension: Complete lack of thematic understanding.

Part 'গ' — Application (প্রয়োগমূলক - 3 Marks)

Cognitive Goal: Mapping characters or themes from the stem (উদ্দীপক) to textbook context (পাঠ্যবই) (Target: 3 distinct paragraphs).

Score	Evaluation Criteria	Teacher Mental Model & AI Instruction
3.0	Para 1: Direct alignment statement. Para 2: Textual context. Para 3: Explicit stem-text synthesis.	Complete Structure: Clear execution across all 3 paragraphs. Exhibits clear knowledge without fluff.
2.0	Textual story explanation is accurate, but the stem-text synthesis in Para 3 is weak or rushed.	Uneven Depth: Student knows the story well but struggles to explicitly connect it to the prompt stem.

Score	Evaluation Criteria	Teacher Mental Model & AI Instruction
1.0	Excessive verbatim copying of the stem text with minimal story reference, or 2 missing paragraphs.	Surface Copying: Student uses stem text to fill space rather than demonstrating knowledge application.
0.0	Unrelated content or missing response.	Unattempted/Irrelevant.

Part 'ঘ' — Higher Ability (উচ্চতর দক্ষতা - 4 Marks)

Cognitive Goal: Independent critical evaluation, synthesis, and original reflection (Target: 4 distinct paragraphs, ~1–1.5 pages).

Score	Evaluation Criteria	Teacher Mental Model & AI Instruction
4.0	Para 1: Clear claim/stance. Para 2: Deep textual analysis. Para 3: Stem-text comparison. Para 4: Personal synthesis (6–8 lines) in student's own words with strong vocabulary.	High Standard: Excellent language expression, correct 4-part structure, precise synthesis, and neat presentation.
3.0	All 4 structural components present; standard explanation; logical conclusion provided.	Solid Baseline: Satisfies all structural and content requirements cleanly without exceptional depth.
2.0	Missing Para 4 (Higher-order synthesis), superficial comparative analysis, or under 1 page in length.	Incomplete Synthesis: Student summarized story events but failed to provide critical analytical evaluation.
1.0	Heavy reliance on verbatim stem text, major structural gaps, or missing paragraphs.	Filler Effort: Low-quality attempt relying heavily on repeating stem text.
0.0	Unrelated or missing response.	Unattempted/Irrelevant.

4. Qualitative Calibration Rules (Non-Binary Soft Weighting)

1. Density Rule (Quality over Quantity) A concise 1-page response in 'ঘ' that correctly executes all 4 structural paragraphs with rich language (ভাষাজ্ঞান) must receive full marks (4/4). Conversely, a 2.5-page response filled with repetitive stem lines must be capped at 2/4.
2. Position Sensitivity & Time-Decay Assessment Evaluators assess each CQ independently. Strong performance in CQ 1 does not grant automatic grace marks for a rushed, low-quality CQ 6 or 7. The AI model must evaluate each question on its own structural merits.
3. Evaluation Context Toggle (Diagnostic vs. Board Mode) • Strict Diagnostic Mode (School/Internal): Strictly enforces headline sentence presence in 'খ' and docks marks for minor structural gaps to encourage learning. • Board Mode (Standardized Exams): Grants partial credit (e.g., 1 mark in 'খ') for incomplete explanations to align with institutional pass-rate requirements.